

Name: _____

UNIT 3: REPRODUCTION

Topic 3: Fertilization & Development

Fertilization

Fertilization is the joining of male and female _____ (sperm and _____) to form a _____ (a fertilized egg) that contains 23 chromosomes from each parent (_____ in total).

➤ *Summarize the four-step process required for fertilization to occur:*

- 1.
- 2.
- 3.
- 4.

Cleavage and Implantation

Name: _____

After fertilization, the zygote continues moving through the oviduct towards the uterus. During this time, several cell divisions occur...



2-cell zygote



4-cell zygote



8-cell zygote



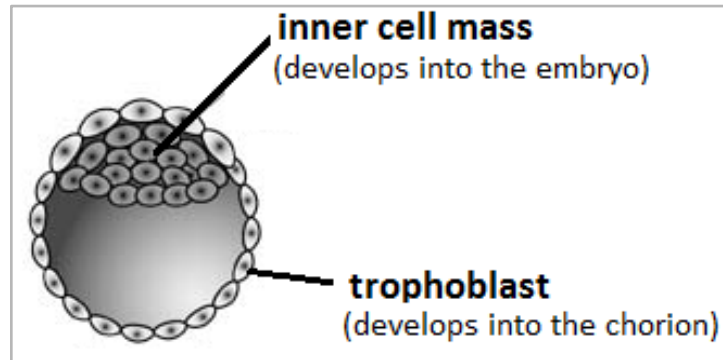
morula

- *In the first few days after fertilization, the zygote undergoes a number of cell divisions referred to as cleavage. What is cleavage?*

- *After the formation of an 8-celled zygote on day five, cleavage occurs several more times to form a sphere of 16 cells. What is this 16-celled sphere referred to as?*

Upon entering the uterus, the morula develops into the _____, a fluid-filled structure consisting of two layers: the _____ and the _____.

Name: _____



➤ What process begins between days 7-14 after fertilization?

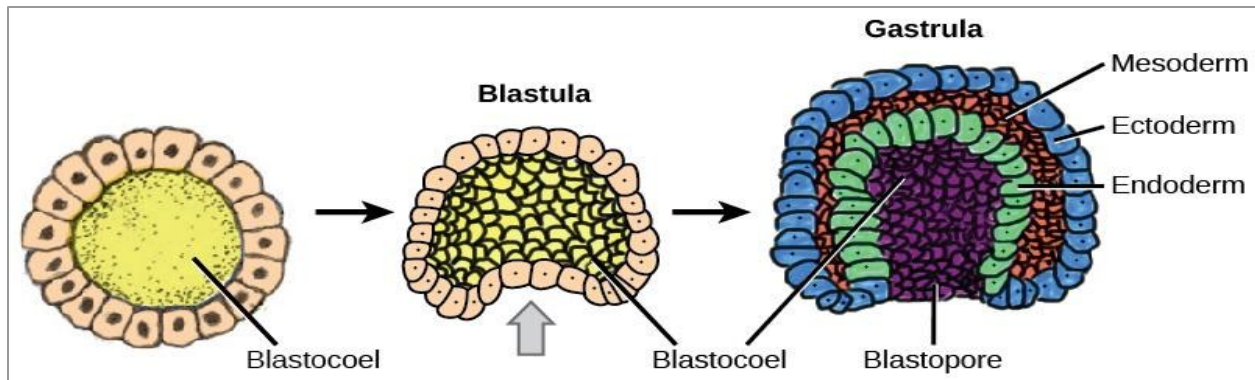
Post-Implantation

- Once implantation occurs, the _____ begins secreting the hormone _____ (hCG).
- hCG maintains the _____ so that it continues to secrete _____ and _____.
- This maintains the _____ and prevents menstruation.
- hCG levels decline after the _____ month of pregnancy, when the _____ is able to secrete these hormones on its own.

Gastrulation

Name: _____

During week 2, the blastocyst begins to change. Through a folding process known as gastrulation, the blastocyst becomes a gastrula, a ~~three~~ ^{three} layered structure that will develop into the embryo...



List the specific tissues and organs that the three distinct germ layers will eventually differentiate into:

- ENDODERM
- MESODERM
- ECTODERM

Organ Formation

Organ formation occurs between weeks 3 & 8, when germ layers begin to fold and early organ systems take shape...

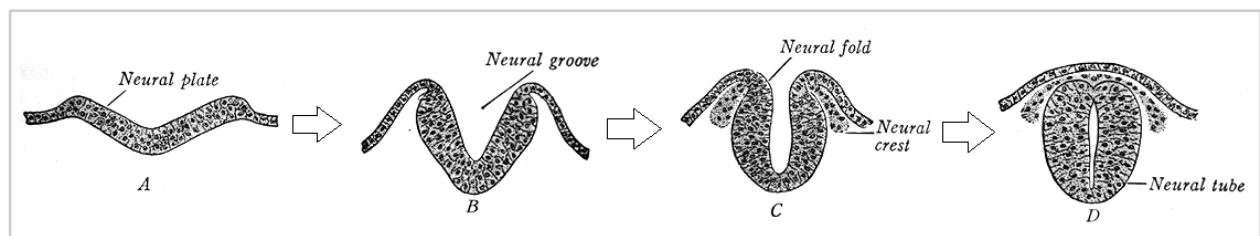
Week	Major Events
3	
4	
5	
6	

Name: _____

7	
8	

Neurulation

Germ layers eventually fold to form the notochord, which will develop into the brain and spinal cord.



➤ *What is required for this process to occur properly?*

Structures that Support the Embryo

Sketch a diagram of a developing embryo (8 weeks in age) in the space below. Label and summarize the function of the allantois, yolk sac, placenta, amnion, and chorion.

A large empty rectangular box provided for the student to draw a diagram of an 8-week developing embryo and label the allantois, yolk sac, placenta, amnion, and chorion.

Name: _____

➤ *What are some other functions of the placenta?*

➤ *The placenta does not filter out teratogens. What is a teratogen? Provide an example.*

Fetal Development

List some characteristics of each of the following trimesters of fetal development.

Trimester	Characteristics
1	

Name: _____

(Weeks 1-12)	
2 (Weeks 13-24)	
3 (Weeks 25-38)	

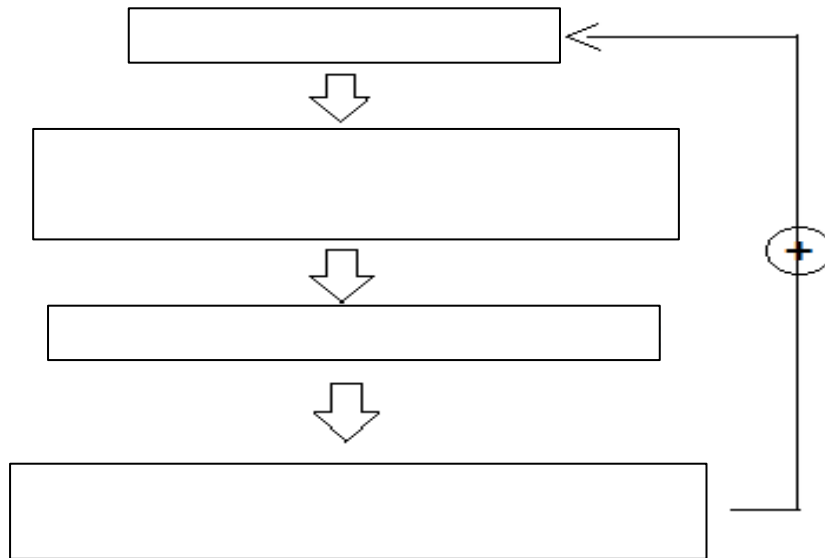
Fetal Growth From 8 to 40 Weeks



Name: _____

Parturition: Delivery of the Baby

The onset of labour, or uterine contractions, is regulated by positive feedback...



- *Contractions are also aided by a group of hormones known as prostaglandins. What are prostaglandins?*

Hormonal Control of Lactation

- After birth, the pituitary begins to secrete _____, which then stimulates _____.

Name: _____

- The actual release of milk from the breasts (a.k.a. _____) is mediated by _____, which causes contractions within the mammary glands.
- Suckling motions send signals through nerve endings to the _____, which then stimulates the pituitary to release _____.